

COMPLETE PROFILE & ARCHITECTURE DOSSIER

Comprehensive System Profile, Historical Ledger & Private Autonomous Agent Blueprint

Subject: Raja Muhammad Ali Asghar (Ali CNC)
Location: Rawalpindi / Sector F-11, Islamabad, Pakistan

Date Generated: September 8, 2026
Status: Complete Intelligence & Architecture Summary

PART 1: PRIMARY IDENTITY & DEMOGRAPHICS

ATTRIBUTE	RECORDED VALUE & HISTORICAL DETAILS
Full Legal / Operating Name	Raja Muhammad Ali Asghar (also known as Muhammad Ali; operates professionally as Ali CNC)
Date of Birth	December 24, 2003 (Age: 22 as of 2026)
Nationality & Residency	Pakistani. Resident of Rawalpindi; maintains workshop operations in Sector F-11, Islamabad.
Core Occupation & Title	2D / 2.5D / 3D CAD/CAM Designer, 3-Axis CNC Router Operator & Programmer, Additive Manufacturing Specialist.
Personality Typology	INTJ (Mastermind / Architect)
Career Origin & Progression	Began professional CNC router file designing and machining on September 7, 2025. Transitions between shops in early 2026 before consolidating operations under Ali CNC and expanding to 3D parametric CAD.

PART 2: PERSONAL RELATIONSHIPS & WORKSHOP NETWORK

INDIVIDUAL / ENTITY	RELATIONSHIP / ROLE	RECORDED CONTEXT & ASSOCIATION
Ghulam Asghar (Raja Ghulam Asghar Abbasi)	Father	Prominent local figure; served as President (Saddar) of the Tajran (Traders Association) in Pindora, Rawalpindi.
Iftikhar (Iftikhar Bhai)	Workshop Mentor & Boss	Senior woodworking and CNC craftsman; regarded as a primary mentor and father figure in precision routing.
Haseeb	Cousin	Close cousin; caretaker of Fargo the German Shepherd dog; subject of transaction and communication tracking.
Fargo	Family Dog (German Shepherd)	German Shepherd cared for alongside Haseeb in Rawalpindi.
R M Faizan	Cousin	Close family relation identified in personal records.
Siblings	Brother & Sister	Immediate family circle recorded in personal profile.
Muneeb, Usama, Merab, Chacha	Workshop Crew	Daily operating team in Sector F-11 workshop (trainees, associates, machine operators, and senior shop coworkers).

PART 3: TECHNICAL SKILLS, TOOLING & HARDWARE ECOSYSTEM

Hardware Machinery & Manufacturing Rigs

- **CNC Router:** Hefei Mingda 1325 (3-Axis Industrial Wood Router), controlled via **NC Studio** motion control interface.
- **Machining Media:** MDF, Lasani, HDF, HDX, super-gloss acrylic/laminate panels, hardwoods.
- **Tooling Inventory:** Flat end mills, V-carve bits (60°, 90°), 6mm ball nose bits (for 3D relief finishing with climb milling).
- **3D Additive Manufacturing:** Bambu Lab A1 3D Printer running Bambu Studio for slicer optimization.
- **Custom Hardware Projects:** Designed an ESP32 / NodeMCU wireless Bluetooth remote pendant communicating with a custom Python background service on Windows to control NC Studio wirelessly.

Software, CAD/CAM & Web Infrastructure

[Vectric Aspire](#)[Onshape 3D](#)[Vector Magic Pro](#)[Adobe Illustrator](#)[KiCad EDA](#)[KeyShot](#)[CAMotics](#)[PTC Creo](#)

[Siemens NX](#)[Fusion 360](#)[AutoCAD](#)[Cloudflare DNS](#)[Vercel](#)[Zoho Mail](#)[DMCA.com](#)

- **Web & Digital Operations:** Fully configured and secured primary web domain `alicnc.pk` via Cloudflare DNS, Vercel deployments, Zoho Mail MX/SPF/DKIM records, and DMCA.com content protection.
- **SVG Format Development:** Developed `encoder.js`, an experimental binary vector format utilizing opcodes, 16-bit delta coordinates, mesh gradients, and raster patches.

Personal Devices & Audio Equipment

- **Mobile Device:** POCO C85 running Xiaomi HyperOS; utilizes Termux, Onshape Mobile, VLC, and MPV.
- **Audio Monitoring Setup:** KZ Castor Pro (Bass Edition) IEMs driven by Conexant CX31993 USB-C DAC dongle; calibrated with Google Sound Amplifier and Poweramp Equalizer custom parametric EQ profiles.
- **Benchmark Audio Interests:** 64 Audio A18t, Sony IER-M500, SoundPEATS T3 Pro.

PART 4: MAJOR PROJECTS & CHRONOLOGY (2025–2026)

DATE / WINDOW	INITIATIVE / APPLICATION	CORE EXECUTION & DETAILS
June 2026	TM-01 Trademark Application (ALI CNC)	Filed official Class 42 trademark application with IPO Pakistan covering CAD, 3D modeling, and engineering design services. Formally responded to examination notices in August 2026 (Application No. 890258).
July 2026	80 Hexagon Interlocking Nesting	Engineered zero-waste puzzle nesting in Vectric Aspire cutting 80 identical interlocking MDF hexagons on a single sheet.
July 2026	TM-01 Trademark Application (AHYEON)	Submitted trademark application with IPO Pakistan under Class 9 for digital media and downloads.
July 2026	Lion Head 3D Relief Carving	Generated 3D mesh model using Hunyuan3D-2; engineered multi-layer slicing and toolpaths for 3-axis CNC router execution.

August 2026	Functional Inverted V-Carve QR Code	Converted, vectorized, and carved a scannable QR code panel in MDF using Vector Magic Pro, Adobe Illustrator, and Vectric Aspire.
August 2026	Municipal Civic Action (CDA / MCI)	Lodged formal civic complaint with Capital Development Authority regarding storm water road flooding outside the Sector F-11 workshop.
September 2026	Y Combinator W27 Application (Forge AI)	Authored application and one-minute founder video script for Forge AI—an AI-driven predictive tool breakage and spindle protection system for CNC routers.

PART 5: PERSONAL INTERESTS, MEDIA & CULTURAL PREFERENCES

CATEGORY	FAVORITES & DEDICATED INTERESTS
K-Pop Ultimate Bias	BIBI (Ultimate bias, admired for genuine engagement) & Jung Ahyeon (정아현) of BABYMONSTER.
Additional Biases & Groups	Jennie (2nd bias), Lisa (original bias), Rosé, Jisoo. Follows BABYMONSTER, BLACKPINK, ILLIT, LE SSERAFIM, TWICE, NewJeans, IVE.
K-Drama Passion	Over 500+ Korean dramas completed. Key favorites include <i>Itaewon Class</i> , <i>Queen Seondeok</i> , <i>The Woman Who Swallowed the Sun</i> . High admiration for actresses Suzy, Jisoo, and IU.
Motorsports & Gaming	Follows Formula 1 Grand Prix racing. Custom scenario and map designer in <i>Age of Mythology</i> .
Streaming Services	Spotify, YouTube Music, Weverse.

PART 6: FULL ARCHITECTURE BLUEPRINT – THE PRIVATE ZERO-DOLLAR AUTONOMOUS AGENT

Summary of the end-to-end private agent system engineered during this discussion to transition away from managed platform boundaries into complete data ownership:

1. The Storage & Data Fabric

- **Supabase PostgreSQL (Free Tier):** Acts as the system brain. Stores prompt templates (agent_prompts), raw chat sessions, parsed notices, and transaction state.
- **PostgreSQL tsvector Full-Text Search:** Provides deterministic lexical search over conversations and knowledge documents with GIN indexing (avoiding hallucinated embeddings for exact part numbers, names, and parameters).
- **Multi-Cloud Storage Hierarchy:**
 - **Cloudinary:** Stores phone screenshots and UI captures; runs on-the-fly image transformations and WebP compression before feeding payloads to Gemini Vision.
 - **Zoho WorkDrive & Google Drive:** Serves as long-term object storage for PDFs, backups, and invoices referenced via Supabase URI pointers.

2. On-Device Android Execution Engine (Zero-Root Architecture)

- **Bridge Layer:** Shizuku + `rish` running locally on Termux under UID 2000 (Shell). Replaces dangerous `su` root shells while granting full hardware UI control.
- **UI Automation:** Employs `uiautomator` `dump` for low-latency XML node hierarchy parsing (eliminating expensive multimodal OCR passes), paired with `input tap`, `input text`, and `screencap -p`.
- **Hardware Access:** Termux API handles direct notifications, SMS interception, battery states, and clipboard reads.
- **Safety Boundary (Human-in-the-Loop):** High-stakes actions (such as money transfers, loan authorizations, or bank interactions) are parsed into a pending ledger state. Execution halts at biometric authentication and system `FLAG_SECURE` boundaries, requiring physical confirmation.

3. Backend Orchestration & The 100% Free Keepalive Matrix

- **Inference Engine:** Google AI Studio Gemini API (Gemini Flash) with direct system instructions and function-calling schemas.
- **Render Web Service Router:** Free-tier container running Express or FastAPI for payload sanitization, failover handling, and long-running pipelines.
- **Keepalive Architecture (Bypassing Inactivity Sleeps):**
 - `cron-job.org` Job A fires an HTTP ping every 10–12 minutes to Render's `/internal/ping` endpoint. The route responds with `200 OK` immediately in-memory (0 DB dependencies) to keep Render awake 24/7 without exceeding the 750 free instance hours.
 - `cron-job.org` Job B triggers a lightweight Supabase query once every 24 hours, completely eliminating the 7-day Supabase inactivity database pause rule.
- **Development Framework:** Autonomous prototyping and artifact generation managed via Google Antigravity 2.0.